

OpenSees Analysis — Full Diagnostic Report

3-Storey Masonry Frame · *newcode3storey.tcl*

0 · What We Know for Certain

The **environment is not the problem**. Running diagnostic.tcl confirmed:

Tcl version	8.6.8
OpenSees version	3.2.1
Macroelement3d	Loaded from external library — Francesco Vanin, EPFL 2019
eleLoad -selfWeight	OK
DisplacementControl	OK
sp in pattern	OK

The errors are in the **analysis logic and model connectivity**, not in the installation.

1 · Reading the Error Output — Line by Line

1.1 · Singular matrix during pushover

```
WARNING BandGenLinLapackSolver::solve() -factorization failed,
      matrix singular U(i,i) = 0, i= 87
WARNING NewtonRaphson::solveCurrentStep() -the LinearSysOfEqn failed in solve()
StaticAnalysis::analyze() - the Algorithm failed at iteration: 0
      with domain at load factor -3.58368e+19
OpenSees > analyze failed, returned: -3 error flag
```

What DOF 87 is:

With 17 nodes × 6 DOFs = 102 total DOFs, and nodes 1 and 2 fixed (12 constrained DOFs), there are 90 free DOFs numbered sequentially. DOF 87 maps to **node 17, DOF 3 (global Z, vertical)**.

Node 17 is the **midpoint internal node of element 9** — the roof spandrel connecting nodes 7 and 8, horizontal along global X, at position [1.5, 0, 9.0].

Node 17 is connected to **exactly one element**. For a horizontal spandrel with local axis along X, the Macroelement3d does not contribute stiffness to the vertical (Z) DOF at the internal node K. That DOF has zero stiffness in the assembled global matrix → zero pivot → singular factorisation.

The same problem likely exists at:

Node 15 (DOF Z) — internal node of element 7 (floor-1 spandrel)
Node 16 (DOF Z) — internal node of element 8 (floor-2 spandrel)
Nodes 9-14 (Y and rotational DOFs) — internal pier nodes, single-element connections

What 'load factor -3.58368e+19' means:

This is **not a structural result**. When the linear solver encounters a zero pivot, it produces a nonsensical displacement. The load factor is computed from that displacement and becomes effectively infinite. This number has no physical meaning — it is a numerical artefact of the singular DOF.

1.2 · Newton fallback and the convergence warnings

```
Newton failed - switching to Newton Initial
WARNING: CTestNormUnbalance::test() - failed to converge but going on
      current Norm: 20394.9 (max: 100, Norm deltaX: 8.43657e-06)
WARNING: CTestNormUnbalance::test() - failed to converge but going on
      current Norm: 9986.6 (max: 100, Norm deltaX: 1.17368e-06)
...
WARNING: CTestNormUnbalance::test() - failed to converge but going on
      current Norm: 596.1 (max: 100, Norm deltaX: 1.82e-07)
```

Why Newton Initial can continue past the singular point:

algorithm Newton -initial uses the initial (undamaged, elastic) stiffness matrix for every iteration instead of updating it. The initial stiffness is non-singular, so the solve succeeds. The trade-off is much slower convergence.

Reading the two norm columns:

Column	Meaning	Values seen
current Norm	Unbalanced force residual (N) at end of step	20 395 → 596
Norm deltaX	Displacement correction at last Newton iteration (m)	$8.4 \times 10^{-6} \rightarrow 1.8 \times 10^{-6}$

The **force residual is large but decreasing step by step** — the system is making progress toward equilibrium. The **displacement correction is near zero** — the DOFs have essentially stopped moving. This is the signature of **post-peak softening with near-zero tangent stiffness**: the structure has passed its lateral capacity peak and is on the descending branch.

What printFlag 5 does:

The **5** at the end of test NormUnbalance 100.0 100 5 means *"print when failed, but return success and continue."* The analysis is NOT stopped — it keeps running and writing output files.

Reliability of results on the softening branch:

With an unbalanced force norm of ~600 N against structural forces in the kN range, per-step errors are roughly 5–20%. The **peak of the pushover curve** — reached before the singular matrix warning — is more reliable. The descending branch should be treated as indicative of trend, not precise quantitative values.

2 · Code Issues — By Confidence Level

2.A · CONFIRMED — Singular DOFs at internal spandrel/pier nodes

Fix: add immediately after the boundary condition block (after fix 2 1 1 1 1 1):

```
# Suppress zero-stiffness DOFs at internal pier nodes
# Fixing Y, Rx, Rz — leaving X, Z, Ry free
foreach n {9 10 11 12 13 14} {
    fix $n 0 1 0 1 0 1
}

# Suppress zero-stiffness DOFs at internal spandrel nodes
# Fixing Y, Z, Rx, Rz — leaving X, Ry free (Z is the critical one)
foreach n {15 16 17} {
    fix $n 0 1 1 1 0 1
}
```

If singular behaviour persists, try fixing all DOFs except X for the spandrel nodes: fix \$n 0 1 1 1 1 1

2.B · CONFIRMED — Pushover fallback overshoots target displacement

When analyze \$nSteps fails at step k , the fallback calls analyze \$nSteps again — running another full set of steps from the current displaced position, overshooting the 0.20 m target.

Replace the entire pushover execution block with:

```
set completedSteps 0
set usingInitial 0

while {$completedSteps < $nSteps} {

    set ok [analyze 1]

    if {$ok == 0} {
        incr completedSteps
        if {$usingInitial} {
            algorithm Newton
            test NormDispIncr 1.0e-6 100 0
        }
    }
}
```

```

        set usingInitial 0
    }
} else {
    if {!$usingInitial} {
        puts "Step $completedSteps: Newton failed – switching to Newton -initial"
        test NormDispIncr 1.0e-5 200 5
        algorithm Newton -initial
        set usingInitial 1
    }
    set ok [analyze 1]
    if {$ok == 0} {
        incr completedSteps
    } else {
        puts "STOP: both algorithms failed at step $completedSteps / $nSteps"
        puts "Roof displacement at stop: [expr $completedSteps * $incr] m"
        break
    }
}
}

if {[expr $completedSteps % 200] == 0 && $completedSteps > 0} {
    puts "Progress: $completedSteps / $nSteps ([expr $completedSteps*$incr] m)"
}
}
puts "Pushover done: $completedSteps / $nSteps steps"

```

2.C · CONFIRMED — Convergence test wrong for post-peak regime

NormUnbalance checks residual forces. In the softening regime the tangent stiffness is near zero, so large force residuals remain even when displacements are fully converged. Use NormDispIncr for the pushover phase.

For gravity (force equilibrium matters):

```
test NormUnbalance 1.0 50 0
```

For pushover (displacement convergence is the meaningful criterion):

```
test NormDispIncr 1.0e-6 100 0
```

2.D · TO VERIFY — Floor load formula

The current formula: `set floorLoad [expr -5.0*$g*$rho*$L_span*$T_pier]`

Dimensional analysis:

ρ [kg/m ³] × g [m/s ²]	= specific weight	[N/m ³]
× L_{span} [m] × T_{pier} [m]	= load intensity	[N/m] (per unit height)
× 5.0	= total force	[N] only if 5.0 is in metres

Numerical result: $-5.0 \times 9.81 \times 1200 \times 3.0 \times 0.25 = -44\,145$ N per floor node

If 5.0 is an equivalent tributary masonry height in metres — the formula is dimensionally consistent and correct.

If 5.0 is a floor pressure in kN/m² — the formula is wrong. The correct line would be:

```
set floorLoad [expr -5.0e3 * $L_span * $T_pier] ;# → -3 750 N per node
```

Please verify what 5.0 physically represents in the original model. This cannot be determined from the script alone.

2.E · RETRACTED — Settlement on fully fixed node

The settlement phase completed without error, and OpenSees 3.2.1 with constraints Transformation appears to handle sp overriding a fix correctly in this version. No change is recommended unless the vertical reaction at node 1 does not change between the gravity and settled states.

3 · Minimum Working Patch (apply in order)

Step 1 — Add after fix 2 1 1 1 1 1 1:

```
foreach n {9 10 11 12 13 14} { fix $n 0 1 0 1 0 1 }
```

```
foreach n {15 16 17} { fix $n 0 1 1 1 0 1 }
```

Step 2 — Change the pushover convergence test:

```
test NormDispIncr 1.0e-6 100 0
```

Step 3 — Replace the `if{$ok != 0}` fallback block with the while-loop from section 2.B.

These three changes together should eliminate the singular matrix warning and the convergence warning flood.

4 · Verifying the Output

After a corrected run, `RoofDisp.out` should contain:

```
time      disp_node8_X
0.0       0.0
0.0001    <small value>
...
0.20      <value near 0.20 if pushover completed>
```

In displacement-controlled pushover, the 'time' column equals the cumulative displacement increment at the controlled node (node 8, DOF 1). It should increase monotonically from 0 to approximately 0.20 m.

Sanity checks on the pushover curve:

1. Initial slope = elastic lateral stiffness.
For this frame: expected range $\sim 10^3 - 10^4$ kN/m.
2. Peak shear per pier $\approx (c + \mu_0 \times \sigma_0) \times L_{\text{pier}} \times T_{\text{pier}}$
where σ_0 = axial stress under gravity loads.
3. If the curve drops sharply right after the elastic range:
 $G_c = 6 \text{ J/m}^2$ is very low. Typical TREMURI values: 20–50 J/m^2 .
Try $G_c = 20$ and compare the descending branch.

5 · Step-by-Step Action Plan

1. Apply the internal node fix (section 3).

Run gravity only (comment out settlement and pushover). Check that it completes without the singular matrix warning.

2. Run gravity + settlement.

Check `SupportReaction.out` — the vertical reaction at node 1 should change between the gravity state and the settled state. If it does not change, release DOF 3 of node 1: `fix 1 1 1 0 1 1`.

3. Run the full script with `NormDispIncr` and the while-loop fallback.

The singular matrix warning should be gone. Some steps near capacity may still use Newton Initial, but convergence warnings should be minimal.

4. Check the floor load (section 2.D).

Verify σ_0 in the piers under gravity. Expected range: 0.5–2 MPa. If much higher, the floor load formula may need correction.

5. Parametric study on G_c .

Try $G_c = 6$ (current), 20, and 50 J/m^2 . Compare the descending branch. This is a valid thesis discussion regardless of the convergence issues.

The structural model and element choice are correct. The issues are in boundary condition completeness (zero-stiffness DOFs at internal nodes), the convergence criterion for the softening regime, and the fallback logic. None of these reflect a misunderstanding of the structural problem.